

CM0832 Elements of Set Theory

List of Axioms

Andrés Sicard-Ramírez

Universidad EAFIT

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Preliminaries

Convention

The number assigned to chapters, examples, exercises, figures, sections, and theorems on these slides correspond to the numbers assigned in the textbook [Enderton 1977].

List of Axioms

Axioms stating the existence of sets

- **Empty (existence) axiom:** There is a set having no members, that is,

$$\exists B \forall x (x \notin B).$$

- **Infinity axiom:** There exists an inductive set, that is,

$$\exists A [\emptyset \in A \wedge \forall a (a \in A \rightarrow a^+ \in A)].$$

List of Axioms

Axioms determining properties of sets

- **Extensionality axiom:** If two sets have exactly the same members, then they are equal, that is,

$$\forall A \forall B [\forall x (x \in A \leftrightarrow x \in B) \rightarrow A = B].$$

- **Regularity (foundation) axiom:** All sets are well-founded, that is,

$$\forall A [A \neq \emptyset \rightarrow \exists m (m \in A \wedge m \cap A = \emptyset)].$$

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Axioms for building sets from other sets

- **Pairing axiom:** For any sets u and v , there is a set having as members just u and v , that is,

$$\forall a \forall b \exists C \forall x (x \in C \leftrightarrow x = a \vee x = b).$$

- **Union axiom** (first version): For any sets a and b , there is a set whose members are those sets belonging either to a or to b (or both), that is,

$$\forall a \forall b \exists B \forall x (x \in B \leftrightarrow x \in a \vee x \in b).$$

- **Union axiom** (final version): For any set A , there exists a set B whose elements are exactly the members of the members of A , that is,

$$\forall A \exists B \forall x [x \in B \leftrightarrow \exists b (x \in b \wedge b \in A)].$$

List of Axioms

Axioms for building sets from other sets (continuation)

- **Power set axiom:** For any set a , there is a set whose members are exactly the subsets of a , that is,

$$\forall a \exists B \forall x (x \in B \leftrightarrow x \subseteq a).$$

- **Subset axiom scheme** (axiom scheme of comprehension or separation): For any propositional function $\varphi(x, t_1, \dots, t_k)$, not containing B , the following is an axiom:

$$\forall t_1 \cdots \forall t_k \forall c \exists B \forall x (x \in B \leftrightarrow x \in c \wedge \varphi(x, t_1, \dots, t_k)).$$

- **Axiom of choice** (a version): For any relation R there is a function $F \subseteq R$ with $\text{dom } F = \text{dom } R$.

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List of Axioms

Axioms for building sets from other sets (continuation)

- **Replacement axiom scheme:** For any propositional function $\varphi(x, y, t_1, \dots, t_k)$, not containing B , the following is an axiom:

$$\forall t_1 \cdots \forall t_k \forall A [\forall x (x \in A \rightarrow \exists! y \varphi(x, y, t_1, \dots, t_k)) \rightarrow \\ \exists B \forall y (y \in B \leftrightarrow \exists x (x \in A \wedge \varphi(x, y, t_1, \dots, t_k)))].$$

References



Enderton, Herbert B. (1977). Elements of Set Theory. Academic Press (cit. on p. 2).